

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Scala for D.S Practical

Practical - 26

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Class : SY-Bsc.IT	Batch : 00
Date of Assignment : 01-09-2026	Date/Time of Submission : 01-09-2026

1. Perform logistic regression using Breeze. Classify a dataset with binary labels.

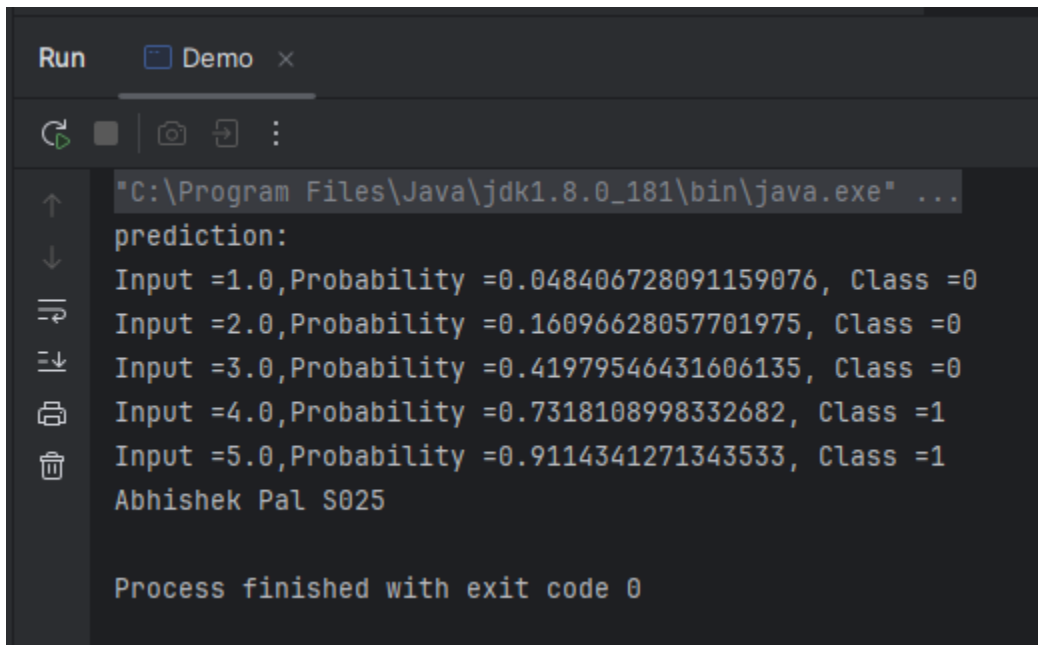
CODE :

```
build.sbt  Demo25.scala x
3  object Demo extends App {
7
8      var weight = 0.0
9      var bias = 0.0
10
11     def sigmoid(z: Double): Double = {
12         1.0/(1.0+ math.exp(-z))
13     }
14     for (i <- 1 to 1000){
15         for (j <- 0 until x.length){
16             val prediction = sigmoid(weight * x(j)+bias)
17             val error = prediction - y(j)
18             weight = weight - 0.02*error*x(j)
19             bias = bias - 0.01*error
20         }
21     }
22     println("prediction:")
23     for(i<-0 until x.length){
24         val probability = sigmoid(weight * x(i)+bias)
25         val result =
26             if (probability >= 0.5)1 else 0
27         println("Input =" +x(i)+",Probability =" + probability +", Class =" +result)
28     }
29     println("Abhishek Pal S025")
30 }
```



```
build.sbt x Demo25.scala
1 ThisBuild / version := "0.1.0-SNAPSHOT"
2
3 ThisBuild / scalaVersion := "2.12.18"
4 libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
5 lazy val root = (project in file("."))
6   .settings(
7     name := "Practical25"
8   )
9
```

OUTPUT :



```
Run Demo x
C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
prediction:
Input =1.0,Probability =0.048406728091159076, Class =0
Input =2.0,Probability =0.16096628057701975, Class =0
Input =3.0,Probability =0.41979546431606135, Class =0
Input =4.0,Probability =0.7318108998332682, Class =1
Input =5.0,Probability =0.9114341271343533, Class =1
Abhishek Pal S025

Process finished with exit code 0
```